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Study on the efficacy of selected biological control agents against *Helicoverpa armigera* infesting chickpea crops

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Abstract- Agricultural productivity is often constrained by insect pests that reduce crop yield and quality. Excessive reliance on synthetic pesticides has resulted in environmental contamination, pest resistance, and loss of beneficial organisms. Biological control offers a sustainable alternative for pest management. The present study evaluated the effectiveness of selected bioagents including *Beauveria bassiana, Metarhizium anisopliae, Trichogramma chilonis, Chrysoperla carnea,* and leaf extract of *Lantana camara* against the gram pod borer, *Helicoverpa armigera*, a major pest of chickpea. Treatments were applied under semi-field conditions, and larval mortality and feeding inhibition were recorded over five weeks. Among all treatments, *B. bassiana* showed the highest larval mortality (79.4%) followed by *M. anisopliae* (71.2%) after five weeks. Egg parasitism by *T. chilonis* reduced larval emergence by 64.8%. Botanical extract caused significant feeding deterrence. Results indicate that integration of microbial, parasitoid, predator, and botanical agents can effectively suppress pest populations and support eco-friendly pest management strategies.

Keywords: *Helicoverpa armigera*, entomopathogenic fungi, parasitoid, botanical extract, sustainable agriculture

INTRODUCTION

Crop protection continues to be a major constraint in achieving sustainable agricultural productivity, particularly in developing countries where pest pressure is high and resource availability is limited. Among the various insect pests, the gram pod borer, *Helicoverpa armigera*, is recognized as one of the most economically damaging polyphagous pests, attacking over 180 cultivated and wild plant species including pulses, cotton, vegetables, and oilseeds.[1,2] In chickpea, larval feeding on reproductive structures such as flowers and pods can result in yield losses ranging from 20% to 80%, depending on infestation severity and environmental conditions.[3]

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For decades, management of *H. armigera* has relied heavily on synthetic chemical insecticides due to their rapid action and ease of application. However, indiscriminate and repeated use of these chemicals has led to several adverse consequences, including the development of insecticide resistance, resurgence of pest populations, contamination of soil and water resources, and detrimental effects on non-target organisms such as pollinators and natural enemies.[4,5] Recent reports also highlight growing resistance of *H. armigera* to multiple classes of insecticides, making chemical control increasingly ineffective and unsustainable.[6]

In this context, biological control has emerged as a key component of integrated pest management (IPM) strategies. Biological control utilizes living organisms such as predators, parasitoids, and microbial pathogens, as well

as plant-derived compounds, to suppress pest populations in an environmentally compatible manner.[7] Entomo pathogenic fungi like *Beauveria bassiana* and *Metarhizium anisopliae* infect insects through the cuticle, proliferate within the host, and cause mortality through toxin production and nutrient depletion.[8,9] Egg parasitoids such as *Trichogramma chilonis* prevent pest establishment by targeting the egg stage, while predators like *Chrysoperla carnea* reduce larval populations through active predation.[10,11] Botanical extracts, including those derived from *Lantana camara*, possess antifeedant, repellent, and growth-disrupting properties that further contribute to pest suppression.[12]

The present study was undertaken to evaluate and compare the efficacy of selected microbial, parasitoid, predator, and botanical control agents against *H. armigera* under semi-field conditions, with the aim of identifying effective combinations for sustainable chickpea pest management.

MATERIALS & METHODS

Study Area

The study was conducted during the rabi season in agricultural fields located in eastern India under semi-field conditions. The region is characterized by a subtropical climate with moderate temperatures and humidity favorable for chickpea cultivation as well as pest proliferation.

Collection of Test Insect

Larvae and egg masses of *H. armigera* were collected from naturally infested chickpea fields. The collected specimens were reared under laboratory conditions (25 ± 2°C and 65–70% RH) on fresh chickpea foliage to maintain a uniform and healthy culture for experimentation.

Methods

Standardized procedures were followed for preparation and application of each biological control agent:

- **Fungal Bioagents:** Conidial suspensions of *B. bassiana* and *M. anisopliae* were prepared from 15-day-old cultures and adjusted to 1 × 10w spores/ml. A surfactant (0.01% Tween-80) was added to enhance adhesion on insect cuticle.
- **Parasitoid Release:** Egg cards containing parasitized eggs of *T. chilonis* were introduced at a standardized release rate equivalent to 50,000 individuals per hectare.
- **Predator Release:** Second instar larvae of *C. carnea* were released at a density of 10 larvae per plant to ensure sufficient predation pressure.
- **Botanical Extract:** A 25% aqueous extract of *L. camara* leaves was prepared and applied as a foliar spray using a hand sprayer.

A completely randomized design (CRD) was followed with six treatments including control, each replicated adequately. Observations on larval mortality, feeding behavior, and developmental changes were recorded weekly for five weeks. Mortality data were corrected using Abbott's formula to account for natural mortality.

RESULTS

All biological treatments significantly reduced larval populations compared to the untreated control, indicating their effectiveness against *H. armigera*. A progressive increase in mortality was observed over time, suggesting cumulative effects of treatments.

Among microbial agents, *B. bassiana* recorded the highest mortality (79.4%) by week five, followed by *M. anisopliae* (71.2%). This indicates superior virulence and adaptability of *B. bassiana* under the given conditions. Predator treatment with *C. carnea* showed rapid initial suppression of early instars (21% in week 1), highlighting its effectiveness in early-stage control, though final mortality (67.6%) was slightly lower than fungal treatments.

The parasitoid *T. chilonis* significantly reduced larval emergence by targeting the egg stage, achieving 64.8% suppression. This demonstrates its importance in preventing population buildup rather than directly killing larvae.

Table 1- Time-wise Variation in Larval Mortality (%) of *Helicoverpa armigera* Across Treatments

Treatment	Week 1	Week 2	Week 3	Week 4	Week 5
B. bassiana	18.20%	34.60%	51.70%	66.30%	79.40%
M. anisopliae	15.90%	29.80%	46.10%	59.20%	71.20%
T. chilonis	12.50%	28.40%	43.70%	56.00%	64.80%
C. carnea	21.00%	37.20%	49.50%	58.40%	67.60%
L. camara extract	10.40%	19.60%	31.20%	44.90%	53.10%
Control	0%	0%	0%	0%	0%

The botanical extract of *L. camara* exhibited comparatively lower mortality (53.1%) but showed strong feeding deterrence and delayed larval development. Such sub-lethal effects are important in reducing crop damage even when immediate mortality is limited.

The results suggest that microbial agents provide higher mortality, while predators and parasitoids contribute to early and preventive control, and botanicals act as supportive agents through behavioral modification.

DISCUSSION

The present findings clearly demonstrate that biological control agents differ in their modes of action and effectiveness against *H. armigera*, supporting earlier observations that multi-trophic approaches are essential for sustainable pest management.[7,13]

The highest efficacy of *B. bassiana* observed in this study is consistent with findings reported by Vega *et al.* (2009)[9], which highlight its strong virulence, ability to penetrate insect cuticle, and production of toxins such as beauvericin that lead to rapid mortality.[8,14] Similarly, *M. anisopliae* has been widely documented as an effective entomopathogen, though slightly less aggressive than *B. bassiana*, which aligns with the comparatively lower mortality recorded in this study.[15]

The performance of *T. chilonis* in reducing egg hatchability corroborates earlier work by Khan *et al.* (2012)[10], which emphasize the importance of egg parasitoids in preventing pest outbreaks rather than controlling established populations.[16] Its role in early-stage suppression makes it a crucial component in integrated strategies.

Predation by *C. carnea* was particularly effective against early instar larvae, supporting findings by Sharma (2005)[3] and subsequent research demonstrating its high feeding capacity and adaptability in field conditions.[11,17] Its effectiveness may decline at later larval stages, explaining the moderate overall mortality observed.

The moderate performance of *L. camara* extract is in agreement with Isman (2006)[12] and recent studies indicating that botanical insecticides primarily act as antifeedants, repellents, or growth regulators rather than direct toxins.[18] Despite lower mortality, their role in reducing feeding damage and delaying development is ecologically significant.

The study reinforces recent research trends advocating integration of multiple biological agents for enhanced pest control efficiency. Combining fungi, parasitoids, predators, and botanicals allows targeting of different life stages of *H. armigera*, reduces pest survival, and minimizes the risk of resistance development.

CONCLUSION

This study shows that biological control agents can effectively manage *Helicoverpa armigera* in chickpea while avoiding the negative impacts associated with chemical pesticides. Among the treatments, the entomopathogenic fungus *Beauveria bassiana* caused the highest larval mortality, indicating its strong potential as a key control agent. *Metarhizium anisopliae* also performed well and can be used alongside other agents to strengthen pest suppression.

The egg parasitoid *Trichogramma chilonis* and the predator *Chrysoperla carnea* played an important role in controlling the pest at its early stages. By targeting eggs and young larvae, they helped limit the buildup of the pest population before serious damage could occur. The leaf extract of *Lantana camara* showed lower direct mortality but reduced feeding and slowed larval development, which contributed to protecting the crop.

Each agent worked in a different way, targeting specific stages of the pest's life cycle. When used together, they provided better control than any single method alone. This kind of combined approach can reduce reliance on synthetic pesticides, support beneficial organisms, and maintain ecological balance in the field.The findings support the use of integrated biological strategies for sustainable chickpea production. Further testing under field conditions and efforts to increase farmer awareness will help in promoting wider adoption of these practices.

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